

Electrical System New Member Projects

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ABSTRACT

This document is intended to provide new members, or anyone who is interested, with the basic knowledge on how to design a basic but still important PCB. It will walk you through how to download Altium Designer and give you two options of basic projects to complete.

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“Any sufficiently advanced technology is indistinguishable from magic.”

- Arthur C. Clarke

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First Things First

Downloading Altium

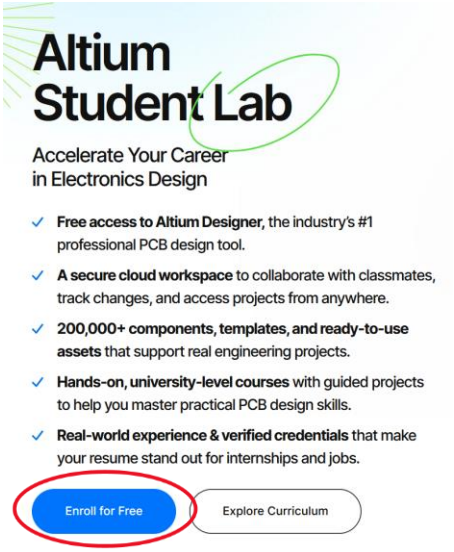
What is Altium Designer?

"Altium Designer is an integrated, user-friendly tool incorporating everything from schematics and layout to simulation and analysis so you can meet design requirements and tight timelines effortlessly."
https://www.altium.com/altium-designer

If you have not heard already, we use Altium to design our boards. This gives everyone a common parts library of components to choose from, Altium has something called the "Manufacturer Parts Search." Altium is the best software you can get for this purpose without having to design your own ECAD (Electrical Computer Aided Design) software. There are free alternatives like KiCAD which are perfectly fine for our purposes as well, but Altium allows for greater project sharing and organization.

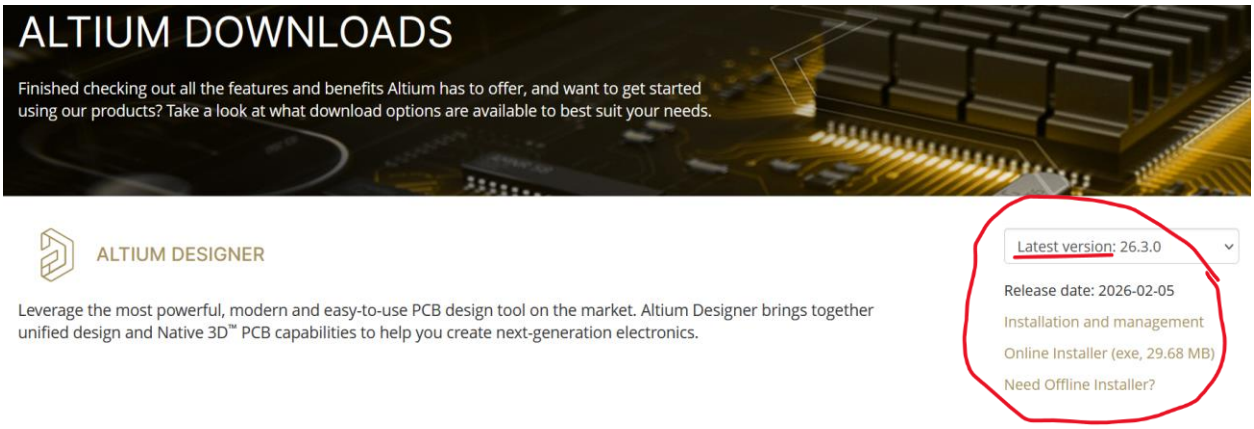
Installing Altium Designer

Altium is nice enough to offer their software for free for students. You do need to use your ufl.edu email address, just go to https://www.altium.com/education/students in order to sign up. It is intimidating software, but do not worry, you will figure it out as you go. Just ask for help if you need it (please do not hesitate to ask for help, we have all been there).

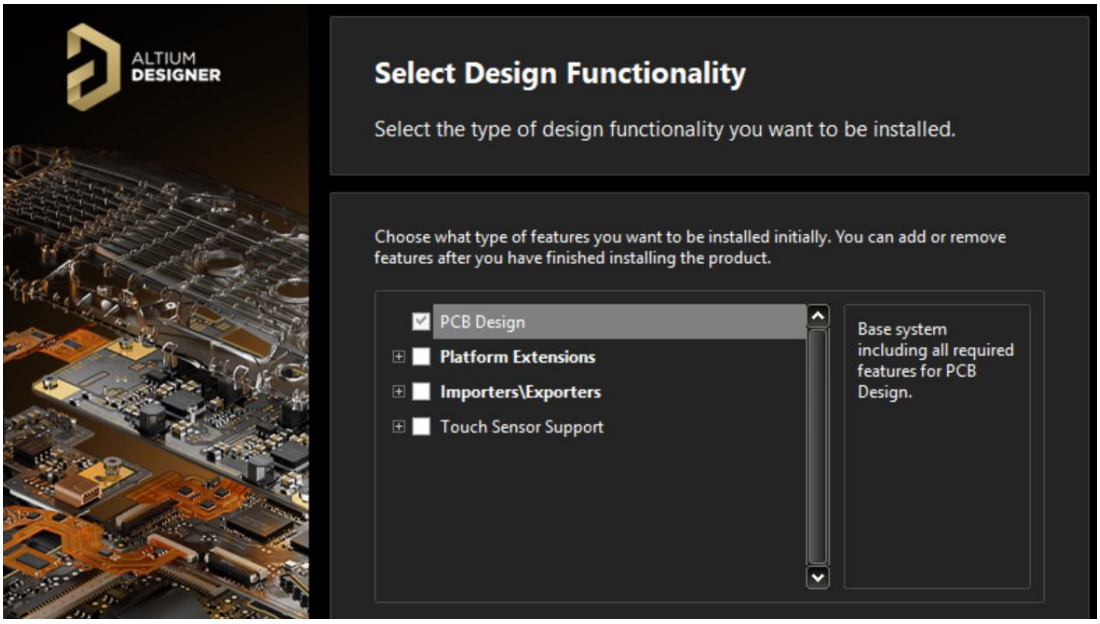


It is also worth going through their learning curriculum to learn how to use the software. As I mentioned before, it is intimidating to use it at first but just watch some tutorials and go through the learning process and you will get there in no time.

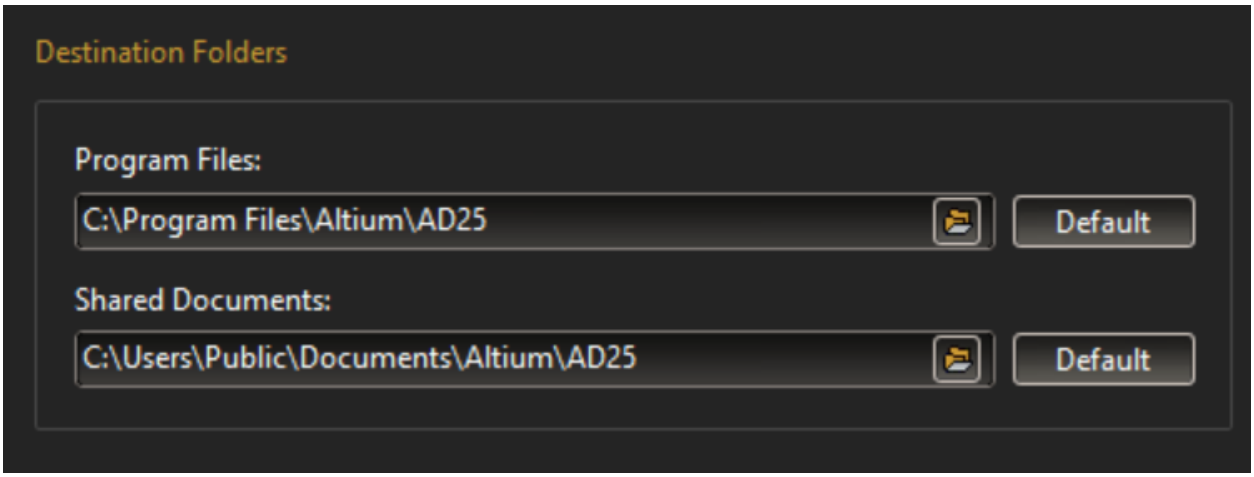
Once you are signed up for Altium, you will need to be added to the Altium 365 workspace, this is where the projects are stored and can be shared between everyone; it is also where the project templates will be. The current Electrical lead will need to add you this before you can start the project. In the meantime, you will need to download Altium Designer, this is the actual program you will use. You can find the application download at [this site (https://www.altium.com/products/downloads)](https://www.altium.com/products/downloads). Make sure to download the latest version.



Feel free to add whatever extensions you want to add; you can always re-run the downloader to add any necessary ones later.



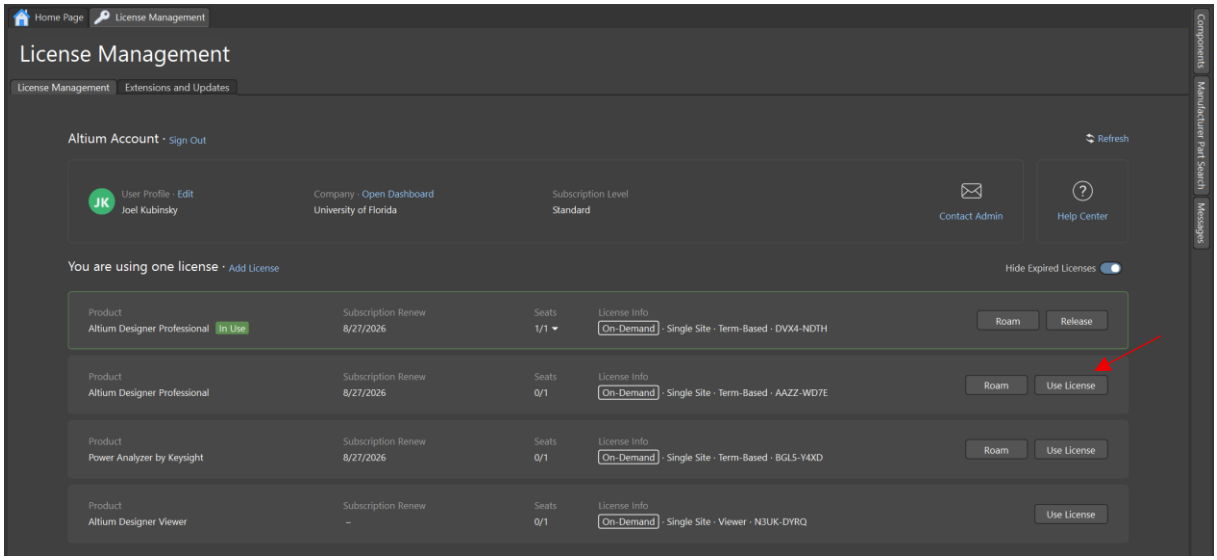
Choose a location you want to store files in; the default is a good place to store files, but it is up to you. This is not a location you want to change later.



You will be asked to sign in during this process and once again when you open Altium Designer for the first time. Soon you will finally be able to get to work though. Once Altium Designer is installed, you will need to switch workspaces. There are a few steps to this, the first of which is activating your license.

Setting up your workspace

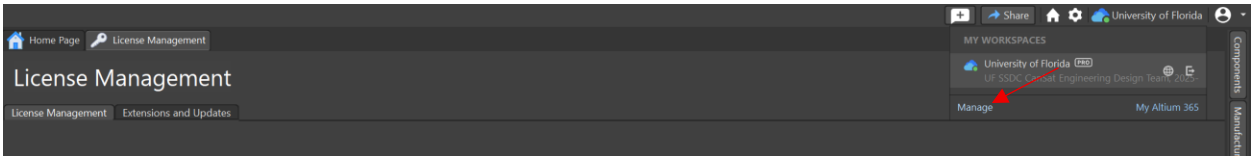




Make sure to select an “Altium Designer” type license.

Once you have chosen a license, there should be a “Choose your workspace” (I unfortunately cannot find the original state of this button, but the location is the same). From there click on “Manage” to add the SSDC workspace. This will take you to an Altium webpage, if you have trouble with this talk to the Electrical lead during one of the team meetings.





Now you will need to choose which project you want to do first. This document will show you how to create a new project for them in each section. Your lead should also be able to walk you through this process if you have any trouble or questions.

Handy notes

It is important to remember that when you are designing circuits, they will exist in real life, so consider their physical properties. For high current traces it may be beneficial to use a polygon pour instead so that the trace impedance can be closer to zero thus reducing any heat created. If you have a differential signal, like USB, they can be what is called an “impedance controlled” signal/trace. This means that their relative lengths can affect signal integrity and may be more susceptible to compromised signal integrity. There is also “cross talk” to consider. This will hopefully make more sense as you go but be sure to research any topics you come across. Most importantly, never be afraid to reach out to your leads and any other members of the team; we learn from each other’s mistakes just as well as we do from our own.

One final note is that you should not be afraid to use both sides of the PCB, the whole point of designing one is the reduce the overall size of the system, by placing resistors and capacitors on both sides of the board you can reduce the footprint of the board overall.

Even More Simple Tutorial

There is an even more simplified tutorial available to show you the basics of Altium Designer. This will show you how to place components, connect them, and the basics of creating a PCB in Altium Designer. I highly recommend at least getting the

documentation for it and looking over it as a reference document as you go through these project boards. Please contact your Electrical Lead if you would like to see this tutorial as well (but it should just be in the Teams as well).

IMPORTANT

Before you move on, it is vital that you remember, and plan how, your boards will be attached to the payload. Including mount holes with a standard size (such as an M3 bolt hole) allows the mechanical teams to easily plan a way to mount them. We need to collaborate with them to allocate space for the boards. Just because we are different teams does not mean we are separate. For as important as it is that the mechanical team does not forget about us, we must **never** forget them as well. Feel free to attend their meetings, talk to the different subsystems inside of mechanical, figure out where your board will go and how you can work with them to easily mount it.

I would also recommend going over the tips I have left at the end of this document, I hope they will help you stay organized and simplify some of the design process.

Power Supply Board

Mission Requirements

The requirements for this board vary from year to year, but usually the competition requires the payload to be able to operate for a **minimum of 2 hours**. This, of course, will depend on the year-to-year. One of the **strict requirements** is that it must **NOT** use a Li-Polymer chemistry. This is for safety reasons as Li-Polymer batteries tend to be quite sensitive and prone to catching fire. And since the payload will land in a, likely dry, grass field, we do not want to burn it down. Please review the mission guide for any additional requirements I may have missed (or were changed) before moving on.

About the Battery Board

The Battery Board is the source of power for the entire payload; thus, it is a critical system. From servos and DC motors to the micro-controller and other sensors, it all starts here. Almost all the components of the payload will use 3.3V or 5V, but other components like servos and DC motors will use the raw output of the battery boards for their supplies.

The actual chemistry may vary year-to-year as well but will usually be a Li-$FePO_4$ (Lithium Iron Phosphate) or Li-NMC (Lithium Nickel Manganese Cobalt Oxide). Whichever chemistry we do end up using, the form factors will be the same. Previously we have used a 2s1p Li-NMC configuration, 2s1p is: 2 in series, 1 in parallel. This is a common way to represent battery configurations, especially as packs get larger (i.e., an EV). We choose this configuration to achieve a higher nominal voltage or ~7.4-8.2V. This allows us to easily use buck converters to drop to 3.3V and 5V power rails, thus giving us slightly higher current as well.

The goal of this project is to reduce the weight of the power board. If you look at the mass budget in the PDR, you will see that the 2 18650 cells we use are ~96g. This may not sound like a lot, but when the total mass budget is 1000g ±10g, those batteries a 10% of our overall weight. This shows that every little bit matters. By switching to 18350 cells, we could potentially cut this figure in half while also freeing up space in an already cramped interior of the payload.

Ideally, you will research and find components to use on your own. This is because as a component ages it can become obsolete, deprecated, or reach an EoL (End of Life) state, and a company may release new versions. But, for the sake of providing a starting direction, a Texas Instruments BQ28Z610 is what we use at the time of creating this document. Feel free to design with this component or find one similar. We do not use all

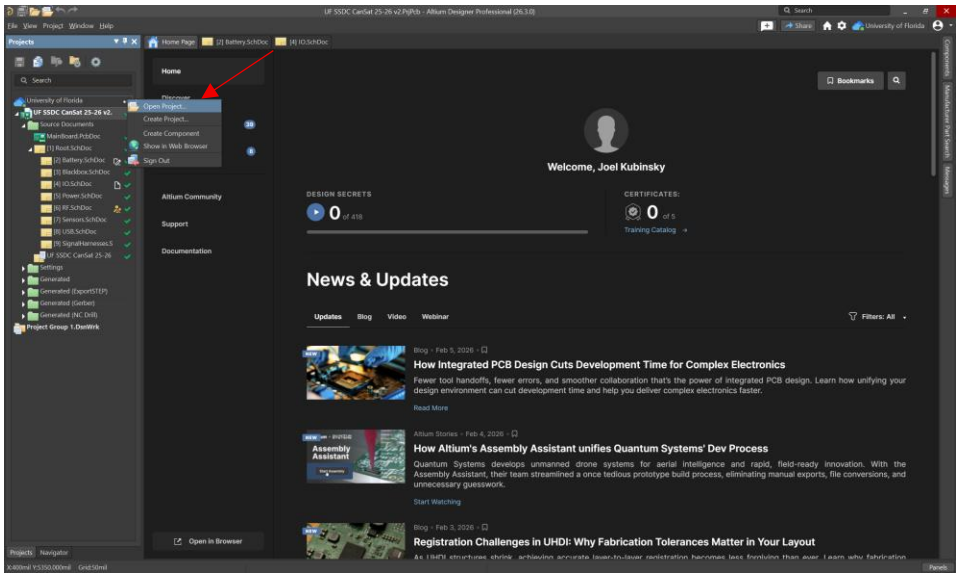
the features of this chip, but it can report all the values we need while also providing battery protection functionality.

Our Requirements

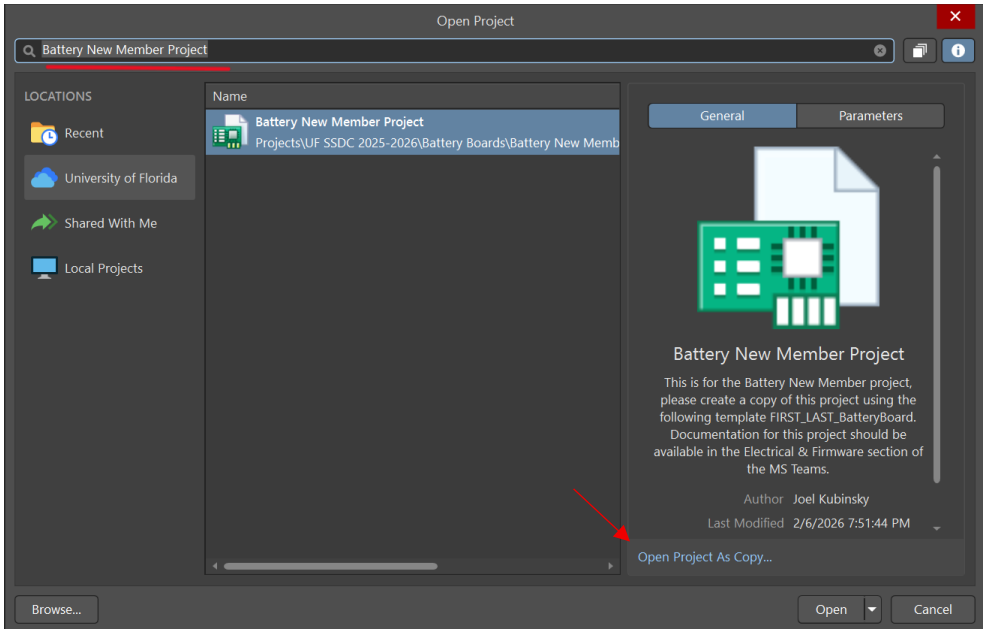
1) The new board should use 2 - 18650 or 18350 cells.
2) The individual cells will need to have their balance managed.
    a. Imbalanced cells can cause voltage issues for our power supplies and can even damage the actual battery cells. This is something that needs to be avoided.
3) Battery voltage and current must be available for monitoring. Coulomb counting also has potential for being required by the competition.
4) The I^2C interface should be used.
    a. Pull-up resistors are not needed on this board and should be placed on the main board.
5) The power board must be able to charge the cells as well.
6) The board should use a 4-pin connector (Molex 70543-0108 or 70541-0073).
    a. The pinout should be as follows:
        i. +BATT (VCC)
        ii. I^2C SCL
        iii. I^2C SDA
        iv. -BATT (GND)
7) There should also be a temperature sensor for battery temperature (i.e., thermistor).
8) These requirements are subject to change as time goes on, and people complete this project and inevitably encounter issues.

Creating your Project

You will create a copy of the project for yourself; this will enable you to experiment with the software while not affecting anyone else or other projects. The first thing to do is open Altium Design and connect to the SSDC workspace. There should be a “Projects” window on the right side of Altium Designer with a cloud icon. Click on the three dots and select “Open Project.”

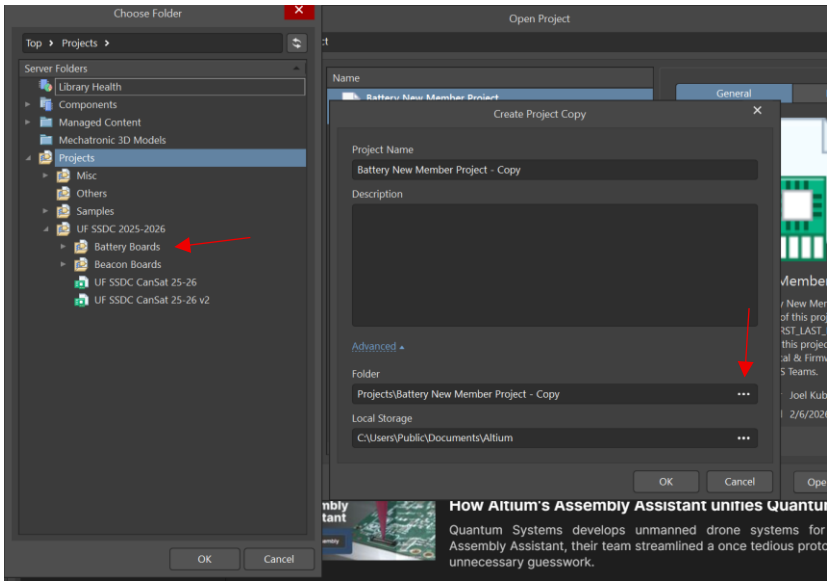


This will open a window on top of Altium, from here you will want to search for the "Battery New Member Project." **DO NOT OPEN THIS PROJECT.** You will want to click "Open Project As Copy..." There are some things to change to, please continue reading before confirming.

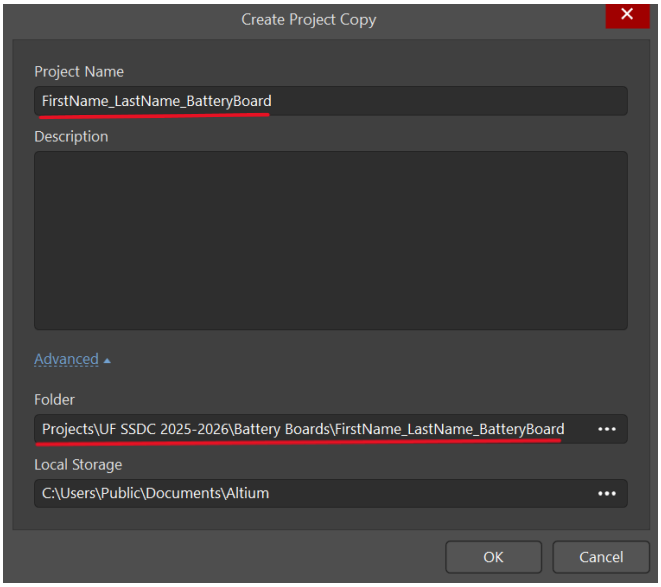


These projects are stored in the Altium 365 workspace, this is like GitHub in a way. It allows anyone to view a project in the workspace without having to open the Altium Designer application. This is useful for quick design reviews or references. To keep the workspace organized, you will need to open the "Advanced" settings and change the "Folder" location, this is **NOT** the "Local Storage" option. Click on the three dots at the end

of the “Folder” entry, this will open another window. Navigate to the “Battery Boards” subfolder as shown.



You will next need to change the “Project Name” field to match the template as shown and as follows: “FirstName_LastName_BatteryBoard.” If the name in the “Folder” does not adjust as well, just repeat the previous step and re-select the “Battery Boards” subfolder again.



Click OK and you can now get started with your new battery board! Goodluck and do not be surprised if you cannot figure something out, just ask for help!

This can also be done more easily by going onto Altium 365, navigating to the project, clicking on its option menu, selecting “Clone”, renaming the project following the

same template, and then just opening the new project in Altium Designer. But where is the fun in that?

Beacon Board

Mission Requirements

As always, when looking to design a new board for the team you must first review the Mission Guide to understand what the system requires. This will help you gain a better understanding of its purpose and should give you a general direction to head in with your design.

About the Buzzer Board

The Beacon Board does exactly what it sounds like it does. It is a self-contained speaker, driver, and battery. It must operate entirely separately from the main board and its power supply. The purpose of the Buzzer is to aid in the recovery of the payload once it has landed. A decent example is to think of a fire alarm, loud enough to want to turn it off but not so loud you would get hearing damage trying to turn it off.

Our Requirements

1) Must be entirely self-sufficient.
    a. This includes its own power supply.
2) It must have an easily accessible power switch.
    a. This can be something as simple as a slide switch, but it must be able to stay on through high G-force events such as the black powder ejection charge and hitting the ground at 5-10 m/sec.
3) There must be a way to adjust the volume of the speaker.
4) It should not be a constant tone.
    a. A low frequency is ideal, potentially one beep lasting 1 second every 5 seconds.
        i. A way to tune this frequency would be a nice plus, but entirely optional.

Creating your Project

You can follow the same steps as outlined in the Battery Board project, just replace any instance of the Battery Board name with the Beacon Board name.

Reminders & Tips

Documentation and Organization

Remember, this is not something you will learn how to do overnight (unless you are told to be the next lead, then you are screwed). You will learn how to do some things better as you go. Do not be afraid to restart a schematic because it simply would take too much effort. The only thing that is more annoying than trying to debug a mystery issue is trying to read the jigsaw puzzle of a schematic that caused it. Good documentation is your friend, and you should try to layout your schematics as best you can.

Symbol Organization

In every component there is a default pin arrangement for their schematic symbols. Often this arrangement is not useful for your purpose and can just add confusion, especially if you use regular wires and not net labels. If this becomes the case, do not be afraid to re-organize the pins represented, or even hide unused ones though I do not recommend it.

Net Labels

In schematics where you might have to run lots of wires around, this is subjective, such as for a series of servos or DC motors and their H-bridges, you may want to use "Net Labels." Net labels are like fancy wires in that they can connect two or more positions on a schematic, they can also be used in signal harnesses to access an individual signal contained within them.

Signal Harnesses

In larger projects you may encounter "Signal Harnesses" Signal harnesses are used to group sets of similar signals together to use less ports in your schematics. This is part keeping schematics clean and organized. If you have more than 2 or 3 all using the same set of 4,5,6 or more signals you may want to use a signal harness. Some teams will even use them for their power rails. Whatever you decide to use, stay consistent in them.

Net Names

Traditionally when describing nets or components that may have a decimal point, such as 3.3V, the unit will be used in place of the decimal point. This is to increase legibility and reduce any mistakes when looking at the schematics. For example: 3.3V would be 3V3, 1.2kΩ would be 1k2R or 1k2Ω while 1.2Ω will be 1R2. Ground is also usually written as

GND. Similarly, in net names you do not use sub-scripts or super-scripts, a signal like I^2C will be written as I2C.

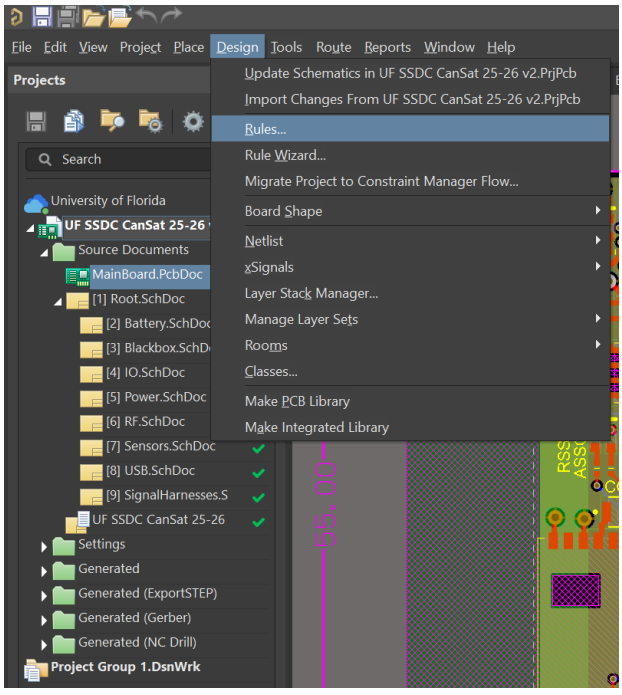
Layer Stackup

Most PCBs designed here will be four layers or less, the typical stack-up will be as follows (Signal1 – Power (VCC) – GND – Signal2) where the Power and GND planes are solid copper fills of just their nets. If you have any RF components, such as a GPS, you may want to swap the Power and GND planes if necessary. When you get into boards with more layers, it is normal to have a GND plane adjacent to every Signal layer. The reason for this is to reduce any parasitic capacitance caused by the gap between signal and ground from the dielectric materials in between them (commonly FR4).

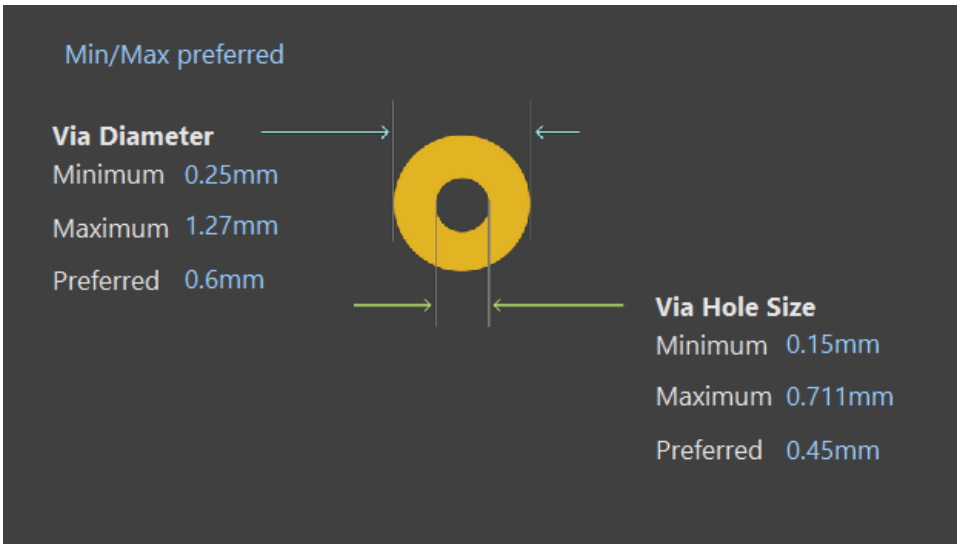
Design Rules

Before you start defining the shape of a PCB, you need to consider that it will need to be manufactured somewhere and that factory will have minimum and maximum capabilities. This can usually be found on their website, but if you have trouble finding or are not sure what Fab to look for, talk to your lead.

To define “Design Rules” you will first need to go into your “.Pcbdoc,” but do not do this until you are done with your schematics as that is more important. Once in the .pcbdoc, go to the “Design” drop down and select the “Rules” menu. This can also be reached by pressing “D” and then “R” short-cuts, look at the underlined character to find its shortcut key.



This will open a menu with a lot of options; it can be a little overwhelming so if you are not sure what to do talk to your Lead and they can walk you through some basic rules. The main categories will be Manufacturing, Electrical, and Routing. Take some time to cross reference the manufacturers' capabilities with those outlined in your Design Rules. This will save you time when trying to layout your design. Here is a sample of a RoutingVias rule set, this defines the minimum, maximum, and preferred size of a via.



The End

If you have any ideas on how to improve this document, talk to your lead. This can be anything from clearer explanations to new ideas for projects for future members to perform.

Stay curious.

"The best design is the simplest one that works" – Albert Einstein